APTM Backend Server (User)

Overview

The **Alpha Phoenix Trading Matrix (APTM)** Backend is a scalable Node.js/Express service that delivers secure *authentication, user profile management, and trading-related features***. MongoDB** serves as the main database, and **JWT** is used for authentication. The server includes advanced data models, document uploads, and user settings management.

Server Architecture

Technology Stack

- **Runtime:** Node.js
- **Framework:** Express.js
- **Database:** MongoDB with Mongoose ODM
- **Authentication:** JWT (JSON Web Tokens)
- **File Upload:** Multer
- **Environment Configuration:** dotenv
- **CORS:** Enabled for cross-origin requests

Project file Structure

```
📁APTM_backend/
├── 📁 config/
│   └── database.js
├── 📁controllers/
│   ├── userController.js
│   └── userProfileController.js
├── 📁 middleware/
│   ├── auth.js
│   └── upload.js
├── 📁models/
│   ├── User.js
│   ├── UserProfile.js
│   ├── UserSettings.js
│   ├── Subscription.js
│   └── UserNotificationPreferences.js
├── 📁 routes/
│   ├── auth.routes.js
│   └── profile.routes.js
├── 📁 uploads/
│   ├── 📁avatars/
│   └── 📁documents/
├── server.js
└── .env
```

API Endpoints

🔐 Authentication Routes – /api/auth

Method	Endpoint	Description	Auth
POST	/signup	Register new user	Public
POST	/verify-otp	Verify email OTP	Public
POST	/resend-otp	Resend OTP	Public
POST	/login	User login	Public
GET	/google	Google OAuth	Public
GET	/google/callback	OAuth callback	Public
GET	/check	Check login state	Protected
POST	/logout	Logout user	Protected
PUT	/password	Update password	Protected
DELETE	/account	Delete account	Protected

👤 Profile Routes – /api/profile

Method	Endpoint	Description	Auth
GET	/me	Get profile	Protected
PUT	/update	Update profile	Protected
POST	/avatar	Upload avatar	Protected
POST	/documents	Upload ID documents	Protected
GET	/settings	Fetch user settings	Protected
PUT	/settings	Update settings	Protected
GET	/subscription	Subscription details	Protected
GET	/public/:username	Public profile	Public

🔧 Utility Routes

Method	Endpoint	Description
GET	/api/test	Health check
GET	/api/debug/routes	List all routes
GET	/api/debug/auth	Debug authentication setup
GET	/api/debug/google-test	Test OAuth config

Data Models

User Model

```
{
  name: String,
  email: String,
  username: String,
  password: String,
  googleId: String,
  isActive: Boolean,
  otpCode: String,
  otpExpiry: Date,
  avatar: String,
  avatarInitial: String
}
```

Extended UserProfile Model

```
{
  userId: ObjectId,
  firstName: String,
  lastName: String,
  username: String,
  bio: String,
  phone: String,
  country: String,
```

```
  timezone: String,
  tradingExperience: [],
  preferredMarkets: [],
  riskAppetite: [],
  interests: [],
  codingExperience: [],
  gender: [],
  idNumber: String,
  dateOfBirth: Date,
  documents: {},
  address: {},
  stats: {},
  verificationStatus: [],
  privacy: {}
}
```

Authentication Flow

1. Email Registration

POST /signup → Create user + OTP

POST /verify-otp → Activate account

POST /login → Issue JWT

2. Google OAuth Flow

GET /google → Redirect to Google

GET /google/callback → Create session

3. JWT Protection

- Requires Authorization: Bearer <token>
- Tokens expire in 24 hours

File Upload System

Supported File Types

- **Avatars:** JPG, PNG, GIF, WebP
- **Documents:** PDF, DOCX, JPG, PNG

Directories

- uploads/avatars/
- uploads/documents/

Environment Variables

PORT=5000

MONGODB_URI=mongodb://localhost:27017/aptm

JWT_SECRET=your-secret

GOOGLE_CLIENT_ID=your-id

GOOGLE_CLIENT_SECRET=your-secret

GOOGLE_REDIRECT_URI=http://localhost:5000/api/auth/google/callback

---

Error Handling

```
{
  success: Boolean,
  message: String,
  user: Object,
  token: String
}
```

Security Features

- Password hashing via bcrypt
- JWT expiration
- File validation & MIME checking
- OTP verification for onboarding
- CORS restrictions

Development Setup

```
npm install
mkdir -p uploads/avatars uploads/documents
cp .env.example .env
npm run dev
```

Test Endpoints

- /api/test
- /api/debug/routes
- /api/debug/auth

Production Deployment

```
NODE_ENV=production
npm run dev
```

Feature Status

Completed

- User auth + OTP
- Google OAuth
- JWT sessions
- Profile & settings management
- File uploads
- Subscription system
- Extended profile fields

In Progress

- Document verification
- Real-time notifications
- Social features
- Trading module integration